

InterviewAce Test Results Summary

Frontend, Backend, AI Scoring, Persistence, Security, and Integration Validation Summary

1. Overview

This document summarizes the final QA execution results for InterviewAce, an AI-powered interview preparation platform built using React, TypeScript, FastAPI, PostgreSQL, sentence-transformers, Ollama, Docker, and GitHub Actions. The purpose of this summary is to document overall testing outcomes, requirement coverage, pass/fail status, integration behavior, and remaining observations identified during validation.

2. Testing Scope Executed

Testing was executed across frontend interaction flows, backend API validation, AI scoring behavior, PostgreSQL persistence, responsive layout validation, deterministic scoring verification, and CI-integrated regression checks. The executed scope included:

- Frontend rendering and interaction testing
- Backend authentication and scoring APIs
- Role and level question retrieval
- AI-generated score and feedback workflows
- AI Coach interaction validation
- Database persistence and restart verification
- JWT protection and security checks
- Mobile responsiveness and UI overflow validation

3. Frontend QA Results

Frontend QA validation confirmed that the InterviewAce user interface successfully supports the complete browser-facing interview practice workflow. Verified frontend behaviors include:

- Role and level selection for SWE, Data Science, PM, and Behavioral tracks
- Question rendering and navigation without full page reloads
- Answer validation for empty, whitespace-only, and short submissions
- Loading states during scoring operations
- Four-axis horizontal score chart rendering
- Responsive behavior at 375 px, 768 px, and desktop widths
- AI Coach panel rendering and message interaction
- History and progress tracking workflows

No critical frontend rendering defects were identified during final validation.

4. Backend QA Results

Backend API validation confirmed that implemented FastAPI endpoints correctly enforce authentication, validation, persistence, scoring, and ownership behavior. Successfully validated backend behaviors include:

- JWT-based authentication and protected endpoint enforcement
- bcrypt password hashing verification
- POST /scoring/submit scoring workflow execution
- Deterministic numeric scoring behavior
- Feedback generation from rubric gaps
- Graceful Ollama fallback handling
- GET /history ownership

protection • PostgreSQL persistence across backend restart • Role-based question retrieval and rubric validation No undocumented endpoints were included in the final validated backend scope.

5. AI Scoring & Feedback Validation

The sentence-transformers scoring pipeline successfully generated four-axis scoring outputs for all validated answer submissions. The following behaviors were confirmed: • technical_depth, clarity, completeness, and structure scores were returned correctly • Score values remained within the expected 0–100 range • Repeated identical answers produced deterministic numeric scores • Rubric-gap feedback remained deterministic across repeated runs • Ollama coaching responses were successfully mocked in CI environments • Live Ollama integration returned valid coaching responses during local integration testing Deterministic numeric scoring requirements were satisfied successfully.

6. Database & Persistence Results

Database validation confirmed that PostgreSQL persistence behaved correctly across answer submission, score storage, and history retrieval workflows. Successfully validated persistence behaviors include: • User records persisted after authentication • Submitted answers linked correctly to user_id and question_id • Score records persisted with all four axis values • Feedback data stored correctly alongside scores • History retrieval returned only authenticated user records • Persisted answers and scores survived backend restart validation • Alembic-managed schemas remained stable during testing No critical persistence or migration failures were identified.

7. Security & Reliability Results

Security testing verified that protected backend routes consistently rejected invalid authentication states. Successfully validated security checks include: • Missing JWT tokens returned HTTP 401 • Tampered JWT tokens returned HTTP 401 • Expired JWT tokens returned HTTP 401 • Passwords were stored as bcrypt hashes • Unauthorized cross-user history access was blocked Reliability testing also confirmed that CI smoke tests, persistence validation, and core integration workflows remained stable during repeated execution.

8. Responsive & Accessibility Results

Responsive validation confirmed that InterviewAce remained usable across mobile, tablet, and desktop viewports. The following responsive requirements passed validation: • No horizontal scrollbars on primary screens • Role and level selectors remained readable on mobile • Horizontal score charts scaled correctly • AI Coach panel remained scrollable and accessible • Text wrapping and overflow handling remained stable • Touch targets met minimum usability sizing expectations No blocking accessibility defects were identified during manual review.

9. Defect Summary

Several minor issues were identified and resolved during QA refinement: • Inconsistent frontend answer length validation • Missing AI Coach frontend coverage • Responsive overflow near charts and labels • Endpoint mismatches between documentation and implementation • Sequence diagram alignment inconsistencies All critical and high-priority issues affecting the core interview practice workflow were resolved prior to final validation.

10. Final QA Assessment

InterviewAce successfully passed frontend, backend, integration, persistence, security, and deterministic scoring validation based on the defined QA strategy, test plans, RTMs, and test case specifications. The platform demonstrated stable behavior across: • Authentication workflows • Role and level question delivery • AI scoring and deterministic feedback • AI Coach integration • Responsive frontend interaction • PostgreSQL persistence and retrieval • JWT security enforcement Final QA assessment: PASS — ready for academic demonstration and submission.

11. QA Execution Summary

Area	Result	Status
Frontend UI & Responsiveness	Validated successfully across desktop/mobile	PASS
Backend API Validation	Authentication, scoring, history, and persistence validated	PASS
AI Scoring & Feedback	Deterministic scoring and rubric feedback confirmed	PASS
AI Coach Integration	Mocked and local Ollama integration validated	PASS
Database Persistence	Answers and scores persisted across restart	PASS
Security Validation	JWT and bcrypt validation passed	PASS
CI / Regression Checks	Smoke tests and builds passed successfully	PASS